

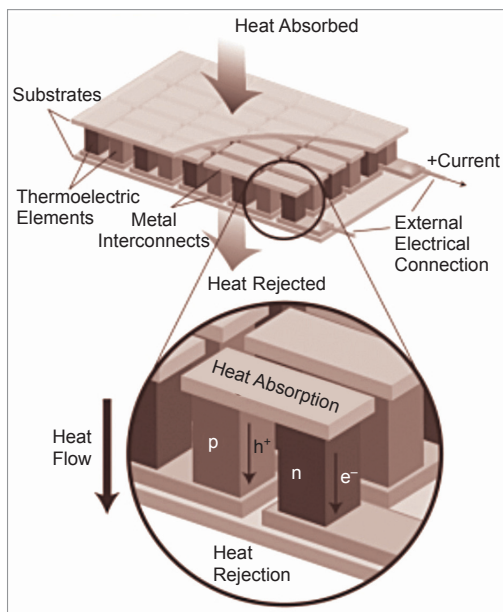


Fig. 5: The concept of thermoelectricity

55nW in standby mode and 10mW in Rx and Tx mode.

The research studies and innovative products given below are those that have emerged recently, showing the true potential of this battery-less technology and its consistent growth.

- Elimination of batteries by using wireless energy transmission is widely seen in Backscatter technology in RFID tags. The RFID reader has a power source which sends the signal. When the tag on the other end receives the energy via electromagnetic waves, its chip is activated when this energy travels through the tag's internal antenna. The remaining energy is modulated with the chip's data and flows back via the tag's antenna to the reader's antenna in the form of electromagnetic waves.

- To increase the safety of the patient, researchers at the Northwestern and George Washington Universities have developed a wireless, battery-free and fully implantable temporary pacemaker that dissolves when no longer needed. The pacemaker harvests energy from external near-field antennas, similar to the RFID technology. This wireless power transfer eliminates any leads that can introduce infections.

- Yet another wireless implantable device is an optoelectronic device for spinal optogenetics (technique that involves the use of light to control

neurons). It consists of a double-layer, rectangular-shaped magnetic coil antenna connected to a microscale inorganic light-emitting diode that can be implanted near the spinal cord for effective simulation of light-sensitive proteins.

- Researchers from the University of Massachusetts Amherst have created a unique solution that can harvest energy from ambient humidity via green materials. The device called Air-Gen, invented in 2020, consists of thin films of protein-nanowires that absorb water vapour from the environment and, due to the conductivity and surface chemistry of the nanowires, creates electricity between the electrodes. The energy produced is enough to

power micro- and nano-electronic devices that can interpret biologically relevant stimuli.

- Researchers at UC San Diego have developed a thin wearable device that can harvest energy from sweat. The wearable device can be wrapped around a fingertip like a band-aid. The padding of carbon foam electrodes absorbs sweat and converts it into electrical energy. The electrodes in the device are equipped with enzymes that trigger chemical reactions between lactate and oxygen molecules in sweat to generate electricity. Underneath the padding is a piezoelectric element that generates additional energy when pressed. This energy is stored inside an integrated capacitor.

Supercapacitors - an alternative to batteries?

Unlike batteries, which rely on electrochemical reaction for charging, supercapacitors rely on electrostatic separation of charged particles in the electrolyte. They can be charged and discharged an infinite number of times until they start bulging.

So, supercapacitors do not require maintenance and are environment friendly, making them last longer as compared to the device they inhabit, while batteries do not. Hence supercapacitors are considered as alternative to batteries. Further advantages of supercapacitors are:

1. Supercapacitors fully discharge with no damage. Capacity is all usable
2. Can be charged and discharged faster
3. They do not swell and shrink in use, and they tolerate ten times the power
4. While charging, no special care is required
5. They do not overcharge and overheat

However, supercapacitors come with their own challenges in terms of energy density, cost, size, and safety, as storing energy is potentially dangerous. But makers are phasing out toxic and flammable chemicals that have been used in supercapacitors, and now they have better safety records than lithium-ion batteries, according to IDTechEx.

Supercapacitors are practically being employed in many large-scale and small-scale applications as an alternative to batteries. A team of electrical engineers at the University of Rochester has developed a supercapacitor based solar harvester for computational systems. The system consists of a microcontroller-driven SEPIC converter that charges supercapacitors and targets embedded system applications requiring power in the 1-10W range.

Talking about the large-scale applications of supercapacitors, Shanghai Aowei buses made by Sinautec Automobile Technologies are fully based on ultracapacitors. These buses rely on supercapacitors for traction, having up to four times the life and greater reliability, though they need top-up charging after every few bus stops.

Conclusion

Looking at the current scenario, most wireless solutions are both battery-free and compact in size. Battery-less solutions, in addition to cost and size benefits, increase safety and reliability. Researchers and R&D engineers across the globe are trying to optimise energy harvesting circuits for more efficiency. Therefore, battery-free solutions will have a big impact on IoT devices, 6G communication systems, and wearables. It is to be seen when this technology will begin to solve major problems of physical batteries. **EFY**

Darshil Patel is a technology enthusiast with a deep interest in Power Electronics